PRODUCT REQUIREMENTS DOCUMENT

# Agentic AI Platforms

Two production-grade product blueprints — warehouse appointment scheduling and B2C travel dynamic packaging.


| AUTHOR | VERSION | STATUS | DATE |
|---|---|---|---|
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# Overview

## DOCUMENT SCOPE

This PRD defines two agentic AI products designed for distinct verticals where autonomous decisioning, real-time orchestration, and deep API integration unlock disproportionate operational value. Each section follows the same structure — executive summary, audience, capabilities, constraints, and KPIs — and is followed by system flow diagrams and an architectural recommendation for the underlying engine.

CONTENTS

PRODUCT 01

# Warehouse Appointment Scheduling

## EXECUTIVE SUMMARY

An intelligent platform designed for Operations Managers to automate, optimize, and manage dock scheduling and appointments. By leveraging agentic AI, the system autonomously negotiates appointment times with carriers, dynamically adjusts schedules based on real-time delays, and optimizes dock door utilization — reducing demurrage fees and increasing warehouse throughput.

> **The wedge:** Most appointment systems are passive calendars. This product replaces the back-and-forth of scheduling emails, phone calls, and yard chaos with an agent that books, negotiates exceptions, and re-optimizes the schedule in real time.

## TARGET AUDIENCE

- **Warehouse Operations Managers** — need visibility and control over dock schedules to allocate labor and equipment.
- **Carriers & Drivers** — require a self-service surface to book, modify, and confirm delivery / pickup slots.
- **Receiving & Shipping Staff** — need accurate, up-to-date schedules to prepare for incoming and outgoing loads.

## CORE CAPABILITIES

| Feature | Description | Priority |
|---|---|---|
| **Autonomous Booking Agent** | AI agent interacts with carriers via email, SMS, or portal to negotiate and confirm appointments based on dock availability and warehouse constraints. | P0 |
| **Dynamic Schedule Optimization** | Dynamically re-allocates dock doors and time slots in response to real-time events — driver delays, early arrivals, extended loading times. | P0 |
| **Predictive Capacity Planning** | ML models forecast dock congestion and labor needs based on historical data, seasonality, and upcoming shipment volumes. | P1 |

| | | |
|---|---|---|
| **Carrier Self-Service Portal** | Secure interface for carriers to view available slots, book appointments, and receive automated confirmations and reminders. | P0 |
| **Automated Exception Handling** | The agent proactively contacts drivers when running late or when warehouse conditions change, negotiating new times without human intervention. | P1 |
| **Integration APIs (TMS / WMS)** | Seamless bidirectional data flow with existing Transportation and Warehouse Management Systems to ensure data consistency. | P0 |

## TECHNICAL CONSTRAINTS

- **Legacy interoperability** — exchange appointment, load, and status events with WMS, TMS, ERP, access control, and yard systems; API scope and event models must be defined early.
- **Operational constraint modeling** — account for door compatibility, labor pools, equipment, temperature / hazmat needs, custom durations, and dynamic buffers driven by historical variability.
- **Resilience & connectivity** — support graceful degradation, queueing, and hybrid deployment for gates and yard zones with inconsistent uptime.
- **Security & access control** — role-based permissions, auditability, and enterprise-grade controls around operational data and integrations.
- **AI observability** — capture traces, tool calls, retrieval context, policy decisions, latency, and quality signals — not just uptime.
- **Human-in-the-loop** — configurable approval thresholds for high-impact actions (priority load reschedules, peak-congestion reassignments, policy exceptions).

## NON-FUNCTIONAL REQUIREMENTS

- Standard booking / rescheduling actions complete in < 5 seconds.
- Multi-site support with site-specific rules, dock taxonomies, and labor calendars.
- Full agent decision history logged for compliance, dispute resolution, and model tuning.
- APIs and webhooks exposed for partner integrations and downstream analytics.

## SUCCESS METRICS

| | |
|---|---|
| **−30%**<br>DRIVER DWELL TIME | **>85%**<br>DOCK UTILIZATION |

**−80%**

MANUAL SCHEDULING TASKS

**−40%**

DETENTION & DEMURRAGE FEES

PRODUCT 02

# B2C Travel Dynamic Packaging

## EXECUTIVE SUMMARY

A B2C-facing intelligent platform that lets travel agencies offer highly personalized, bundled itineraries — flights, hotels, activities, transfers — in real time. The AI acts as a personal concierge, conversing with the customer to understand preferences, budget, and constraints, then autonomously queries multiple GDS and aggregators to construct and price the optimal package.

> **The wedge:** Static packages convert poorly because they ignore intent. A conversational agent that assembles inventory on-demand turns vague trip dreams into bookable, margin-optimized bundles in a single session.

## TARGET AUDIENCE

- **B2C Travelers** — seeking frictionless, personalized vacation planning without stitching components together themselves.
- **Travel Agency Owners** — looking to increase margins through opaque bundled pricing and lift conversion via hyper-personalization.

## CORE CAPABILITIES

| Feature | Description | Priority |
|---|---|---|
| **Conversational Concierge** | NLP-driven agentic interface (chat / voice) that elicits user preferences and refines options interactively. | P0 |
| **Dynamic Packaging Engine** | Simultaneously queries flights, accommodations, and ancillaries across GDS and bedbanks to assemble a single unified itinerary. | P0 |
| **Opaque Pricing & Margin Optimization** | Dynamically prices the bundle to maximize agency margin while remaining competitive — hiding individual component costs. | P1 |
| **Context-Aware Recommendations** | ML suggests relevant add-ons (tours, insurance, car rentals) based on user profile, destination, and current bundle composition. | P1 |

| | | |
|---|---|---|
| **Automated Itinerary Management** | Agent monitors booked components for disruptions and autonomously proposes or executes rebooking alternatives. | P2 |
| **Multi-Source Inventory API** | Robust integrations aggregating real-time pricing and availability from flights, hotels, and activity suppliers. | P0 |

## SUCCESS METRICS

**+20%**
BOOKING CONVERSION RATE

**+15%**
AVERAGE ORDER VALUE

**−50%**
SEARCH-TO-BOOK TIME

**+10%**
AGENCY PROFIT MARGIN

AUTHOR ALIGNMENT

# Why this author, why these products

## Xavier Mascaros · strategic fit

- **AI/ML strategy & definition.** Track record of defining product visions and prioritizing ML use cases (intelligent shipment routing at CargoSprint, Trust & Safety ML at Twitter). This PRD frames AI as a driver of core KPIs — not a technology demo.
- **API ecosystems & integration.** Both products lean on robust API ecosystems (TMS / WMS for warehouse, GDS / bedbanks for travel). Experience growing enterprise API platforms at Accelya and managing integrations at UKG maps directly.
- **Supply chain & logistics depth.** Senior Software Consultant at Manhattan Associates — drove SDLC implementations for Fortune 500 supply chain clients. The domain knowledge for the warehouse product is in hand.
- **Airline & travel tech.** Tenure at Accelya managing the airline distribution roadmap (GDS agencies, conditional pricing logic) translates directly to the dynamic packaging engine.

# System Flow Diagrams

## Warehouse Appointment Agent

| | | |
|---|---|---|
| Carrier → | AI Booking Agent | "Request delivery slot" |
| Agent → | WMS / TMS APIs | Query load + constraints |
| WMS → | Agent | Cold storage · 2h dwell |
| Agent → | Dock Schedule | Check dock + labor capacity |
| Dock → | Agent | Available slots |
| Agent → | Carrier | Propose optimal slots |
| Carrier → | Agent | Confirm selection |
| Agent → | Dock | Lock appointment |
| Agent → | WMS | Update inbound timeline |
| Agent → | Carrier | Confirmation + QR code |

## Travel Dynamic Packaging Agent

| | | |
|---|---|---|
| Customer → | AI Travel Concierge | Preferences + budget |
| Concierge → | Packaging Engine | Structured intent |
| Engine → | GDS / Bedbanks | Query flights · hotels · ancillaries |
| GDS → | Engine | Inventory + net prices |
| Engine → | Engine | Apply opaque pricing + margin |
| Engine → | Concierge | Unified bundled options |
| Concierge → | Customer | Personalized package |
| Customer → | Concierge | Confirm + pay |
| Concierge → | Engine | Execute transaction |
| Engine → | GDS | Secure inventory · issue tickets |
| Engine → | Customer | Final itinerary |

SECTION 05

# Engine Build Strategy

Constructing the core agentic engine for both platforms forces a critical architectural decision. Three paths considered; one recommended.

### OPTION A · FULLY CUSTOM BUILD

Open-source LLMs (e.g. Llama 3) with custom orchestration built from scratch.

**PROS**

- Complete data privacy
- No vendor lock-in
- Highly customizable agent loops
- Zero recurring token API costs at scale

**CONS**

- Extremely high upfront R&D cost
- Requires specialized ML engineering talent
- Significant infrastructure overhead
- Maintenance burden for tool-calling updates

### OPTION B · PLATFORM-AS-A-SERVICE

Managed services like AWS Bedrock Agents, Azure OpenAI Assistants, or Vertex AI.

**PROS**

- Extremely fast time-to-market
- Built-in tool calling, state, and memory
- Out-of-the-box enterprise security
- Continuous upstream model improvements

**CONS**

- High API costs as volume scales
- Ecosystem lock-in
- Less control over the orchestration loop
- Potential latency on shared platforms

### OPTION C · HYBRID ARCHITECTURE RECOMMENDED

A managed LLM for complex reasoning and NL generation, paired with an in-house orchestration layer (LangGraph / AutoGen) and proprietary API integrations.

**PROS**

- Balances rapid speed-to-market with deep architectural control

**CONS**

- Requires diligent API abstraction design upfront to prevent tight

- Orchestration logic — the "brain" connecting to WMS or GDS — remains proprietary IP
- Allows swapping foundational LLMs via API abstraction as cost / performance shifts

coupling to a single foundational model provider